

LONG MEMORY

$$\gamma(k) = \text{cov}(X_j, X_{j+k}) = \text{cov}(X_0, X_k)$$

SHORT MEMORY

$$\sum_{k \in \mathbb{Z}} |\gamma(k)| < \infty \quad \text{and} \quad \sum_{k \in \mathbb{Z}} \gamma(k) > 0$$

LONG MEMORY

$$\sum_{k \in \mathbb{Z}} |\gamma(k)| = \infty$$

NEGATIVE MEMORY

$$\sum_{k \in \mathbb{Z}} |\gamma(k)| < \infty \quad \text{and} \quad \sum_{k \in \mathbb{Z}} \gamma(k) = 0$$

MODEL

CONTINUOUS TIME

$$V_T(\phi) = - \int_0^T \phi_u S_u du - \lambda \int_0^T |\phi_u|^\alpha du$$

DISCRETE TIME

$$V_T(\phi) = - \sum_{k=0}^T \phi_k S_k - \lambda \sum_{k=0}^T |\phi_k|^\alpha$$

MARKET BOUND

$$V_T(\phi) \leq C \sum_{k=0}^T E |S_t|^{\frac{\alpha}{\alpha-1}}$$

OPTIMAL GROWTH

VALUE FUNCTION

$$u(T) = \sup\{E[V_T(\phi)], \phi \in \mathcal{X}_T\}$$

PROBLEM

Find ϕ^* such that $E[V_T(\phi^*)]$ has the same growth as $u(T)$.

RESULT ON LONG MEMORY

$$\gamma(k) = \text{cov}(S_1 - S_0, S_{k+1} - S_k)$$

ASSUMPTION:

Let $H \in [1/2, 1]$. The process S is centered Gaussian with stationary increments and

$$\text{cov}(S_u - S_t, S_t) \geq 0$$

and for some constants $J_1, J_2 > 0$

$$\gamma(0) = 1, \quad k^{2H-2} \leq \gamma(k) \leq k^{2H-2}, \quad \text{for } k \geq 1.$$

RESULT ON LONG MEMORY

STRATEGY:

$$\phi_t^* = \phi_t^*(T, \kappa) = \begin{cases} \text{sgn}(S_t)|S_t|^\kappa & t \in [0, T/2] \\ -\frac{1}{T/2} \int_0^{T/2} \phi_s^* ds & t \in [T/2, T] \end{cases}$$

THEOREM: With the ASSUMPTION in force we have

- * Maximal expected profits satisfy

$$\limsup_{T \rightarrow \infty} \frac{u(T)}{T^{H(1+1/(\alpha-1))+1}} < \infty$$

- * For each $0 < \kappa < \frac{1}{\alpha-1}$ the strategy ϕ^* satisfies

$$\lim_{T \rightarrow \infty} \frac{EV_T(\phi^*)}{T^{H(1+\kappa)+1}} > 0$$

RESULT ON NEGATIVE MEMORY

$$\gamma(k) = \text{cov}(Z_0, Z_k),$$

where the Z_k is defined below.

ASSUMPTION:

Let $H \in [0, 1/2]$. Let Z_t , $t \in \mathbb{Z}$ be a real-valued, zero mean, stationary Gaussian process which will represent price increments. Let $\gamma(k) = \text{cov}(Z_0, Z_t)$, $t \in \mathbb{Z}$ denote its covariance function. We assume that there exist $T_0 > 0$ and $J_1, J_2 < 0$ such that for all $t \geq T_0$

$$J_1 t^{2H-2} \leq \gamma(k) \leq J_2 t^{2H-2}$$

RESULT ON NEGATIVE MEMORY

STRATEGY:

$$\phi_t^* = \phi_t^*(T, \kappa) = \begin{cases} -\text{sgn}(S_t)|S_k|^\kappa & k \in [0, T/2] \\ -\frac{1}{T/2} \sum_{k=0}^{T/2} \phi_k^* & k \in [T/2, T] \end{cases}$$

THEOREM: With the ASSUMPTION in force we have

- * Maximal expected profits satisfy

$$\limsup_{T \rightarrow \infty} \frac{u(T)}{T^{H(1+1/(\alpha-1))+1}} < \infty$$

- * the strategy ϕ^* satisfies

$$\lim_{T \rightarrow \infty} \frac{EV_T(\phi^*)}{T^{H(1+\frac{1}{\alpha-1})+1}} > 0$$